

Topic 6: Conditional Statements

5.1 What is Conditional Logic?

- Programs need to make decisions based on data
- Without conditionals, a program runs the same steps every time regardless of input
- C++ provides: `if`, `else if`, `else`, and `switch`
- Indentation is not required by the compiler but is essential for readability

5.2 if and else

```
int marks = 75;
if (marks >= 50) {
    cout << "Pass" << endl;
} else {
    cout << "Fail" << endl;
}
```

- Condition inside parentheses after `if`
- Curly braces `{}` contain the block to execute
- `else` runs when the `if` condition is false

5.3 else if for Multiple Conditions

```
if (marks >= 80) cout << "A";
else if (marks >= 70) cout << "B";
else if (marks >= 60) cout << "C";
else if (marks >= 50) cout << "D";
else cout << "F";
```

- Conditions are checked top to bottom — first true one executes, rest are skipped
- Can omit braces for single-statement blocks (but using braces is safer habit)

5.4 The switch Statement

```
int choice = 2;
switch (choice) {
    case 1: cout << "New Game"; break;
```

```
case 2: cout << "Load Game"; break;
case 3: cout << "Exit"; break;
default: cout << "Invalid choice";
}
```

- switch tests one variable against multiple exact values
- `break` is required to prevent fall-through to the next case
- `default` runs if no case matches — equivalent to `else`
- switch only works with int, char, and enum — not float or string

5.5 if-else vs switch

if-else	switch
Any condition	Exact value comparison only
Works with ranges	Works with int/char/enum
More flexible	Cleaner for many exact values

5.6 Decision Flowchart

- If-else logic maps directly to a flowchart: diamond = decision, two arrows = true/false
 - Drawing the flowchart first helps plan the code
 - Nested if = nested diamonds in the flowchart
-